

International Journal of Human–Computer Interaction

ISSN: 1044-7318 (Print) 1532-7590 (Online) Journal homepage: www.tandfonline.com/journals/hihc20

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To cite this article: Eunsun Cho & Jong-Youn Rha (2026) Does Consumer Autonomy Matter? Consumer Perceptions of AI-Driven Recommendations and the Mediating Role of Consumer Autonomy in Consumer Satisfaction, International Journal of Human–Computer Interaction, 42:14, 10949-10971, DOI: 10.1080/10447318.2025.2587927

To link to this article: https://doi.org/10.1080/10447318.2025.2587927

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INTERNATIONAL JOURNAL OF HUMAN–COMPUTER INTERACTION
2026, VOL. 42, NO. 14, 10949–10971
https://doi.org/10.1080/10447318.2025.2587927

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Does Consumer Autonomy Matter? Consumer Perceptions of AI-Driven Recommendations and the Mediating Role of Consumer Autonomy in Consumer Satisfaction

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ABSTRACT

AI-driven recommendations help consumers efficiently find desired products, yet they can also act as external influences that intervene in consumer decision making. In the era of AI, it is essential for consumers to have opportunities to reflect their own values and desires, which is crucial for consumer-oriented AI technologies. This study examines the mediating role of consumer autonomy in the relationship between consumers' perceptions of AI-driven recommendation systems and their satisfaction. We propose that positive and negative perceptions influence satisfaction through consumer autonomy. In-depth interviews revealed ambivalent views regarding profiling for data input, algorithm as throughput, and recommended alternatives as output. Structural equation modeling showed that "alternative discovery," "personalization," and "manipulative practices" partially mediated consumer autonomy and satisfaction, while "information bias" and "privacy intrusiveness" were fully mediated by consumer autonomy. Consumer autonomy emerged as the most significant predictor of satisfaction, underscoring its critical role in shaping AI shopping experiences.

KEYWORDS

AI-driven recommendations; consumer autonomy; consumer perception; satisfaction with shopping experience; mediation effect

1. Introduction

In the retail environment, AI-driven recommendations drive consumer satisfaction. AI systems analyze individuals' online activities, preferences, and search histories to recommend suitable products, making it easier for consumers to find items that align with their preferences. Consumers can discover new interests or needs through AI-driven recommendations, thereby facilitating efficient decision making (Wertenbroch et al., 2020). These systems are evolving into hyper-personalization by analyzing consumers' current situations, needs, and behavioral patterns in real time. Thus, AI-driven recommendations are leading innovative changes in the retail environment, offering new opportunities for both consumers and sellers.

Despite the advantages, there are concerns that the expanding use of AI-driven recommendations may promote information bias and pose risks to personal privacy (Wertenbroch et al., 2020). Consumers also find it difficult to understand the black-box algorithms (Berente et al., 2021) and marketing tactics such as micro-targeting or behavioral nudges used in hyper-personalization strategies, which may threaten consumer autonomy (Hyman et al., 2023). Although consumers appreciate the convenience and personalized services offered by AI technology, they also have concerns about the underlying issues.

The importance of human-centered development to ensure a safe AI society has been highlighted. The UNESCO guidelines on AI ethics emphasize that human rights and fundamental freedoms must

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Supplemental data for this article can be accessed online at https://doi.org/10.1080/10447318.2025.2587927.

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be respected, protected, and enhanced throughout the lifecycle of AI systems (UNESCO, 2021). Given that human freedom and rights include the exercise of autonomy, it is necessary to examine whether technology is evolving in a direction that can realize consumer autonomy as advised by these guidelines.

Consumer autonomy is fundamental to consumer sovereignty because autonomous consumer choices significantly impact market resource allocation. Therefore, it is crucial to ensure consumer autonomy in environments with AI-driven recommendations to foster a fair market environment. In AI-based environments, consumer autonomy can be compromised when businesses use covert influence strategies, often without consumer awareness (Bjørlo et al., 2021). However, as consumers gain better access to quality information, they also potentially gain more autonomy.

Prior research has primarily focused on theoretical discussions about preserving consumer autonomy (André et al., 2018; Bjørlo et al., 2021; Fassiaux, 2023; Hyman et al., 2023; Wertenbroch et al., 2020). Although some empirical studies have been conducted (Flacandi et al., 2024; Sankaran et al., 2021; Xu et al., 2020), little research has comprehensively considered diverse consumer perceptions to understand what factors affect consumer autonomy and satisfaction.

Tailored services for AI-driven recommendations are generally perceived positively as they offer numerous benefits to consumers. However, as AI-driven recommendations become more common, researchers are concerned that negative perceptions could degrade satisfaction. AI technology involves complex characteristics, incorporating both positive and negative responses (Yap et al., 2021), thus necessitating a multifaceted examination of consumer perceptions. Balancing these perceptions in the context of technological advancement is essential to understand societal changes due to technology and to prepare for appropriate technological, social, and policy responses.

While previous studies have primarily focused on identifying direct relationships between consumer perceptions and satisfaction, this study introduces consumer autonomy as a psychological mechanism that links perceptions to satisfaction. By empirically testing the mediating role of autonomy, the study provides a more nuanced theoretical framework for understanding consumer experiences in AI shopping environments.

Prior research has identified a range of positive consumer perceptions of AI-driven recommendation systems, such as efficiency, personalization, and enjoyment, that can enhance user satisfaction (Noor et al., 2022; Yin & Qiu, 2021). In contrast, negative perceptions have largely focused on privacy concerns, especially within the personalization–privacy paradox framework (Wang et al., 2024). This study broadens the scope by empirically incorporating additional ethical concerns such as algorithmic opacity, information bias, and manipulative practices. These factors have often been addressed only conceptually in previous research (Bjørlo et al., 2021; Wertenbroch et al., 2020). Thus, the current study highlights critical elements for designing consumer-centric AI recommender systems, incorporating both positive and negative factors.

To identify these perception dimensions, this research first adopts a qualitative approach. Given that AI systems may elicit unfamiliar psychological responses, it is essential to explore consumer experiences inductively before developing the measurement scale. This two-stage design allows the model to reflect the complexity of real-world user experiences rather than imposing predefined assumptions derived from unrelated contexts.

This study posits that consumers' positive perceptions of AI-driven recommendations enhance consumer satisfaction by increasing the potential for autonomy in making free purchase decisions. Conversely, negative perceptions, which have typically been dismissed as detrimental, essentially reduce the possibility of realizing autonomy, thereby impeding satisfaction. This research reports that both positive and negative perceptions simultaneously impact satisfaction. Thus, we seek to explain the psychological mechanism mediating the relationship between consumer perceptions and satisfaction through consumer autonomy. The research objectives are as follows: (1) to examine both positive and negative consumer perceptions influencing consumers' satisfaction with AI-driven recommendations, and (2) to verify how consumer autonomy mediates this relationship. The aim is to leverage the findings to foster discussions on how to enhance consumer satisfaction in the AI market environment without losing the vital virtue of consumer autonomy.

2. Theoretical background

2.1. Consumer autonomy

Autonomy originates from the Greek words "auto" (self) and "nomos" (rule, law), signifying that an individual's decisions and actions are inherently their own (De Tavernier, 2012; Smith et al., 2013). From a philosophical perspective, autonomy is a crucial concept associated with realizing a subjective human life. Individual autonomy and free will are central to Western Enlightenment thinking and provide the foundation for economic theories regarding consumer choice (Wertenbroch et al., 2020). Personal choice has long held a central position in theories of modern capitalist economies. For neoliberal economists, freedom of individual choice is regarded as an inalienable right (Dholakia et al., 2021). The ability to choose is often likened to democracy, where the market operates as a "democracy of goods" or "consumer democracy," with consumer choices acting as a form of "voting" that determines the fate of products and companies (Schwarzkopf, 2011, p. 8).

In the AI-based market, consumer autonomy signifies perceived autonomy, encompassing the awareness and acceptance of external influences and representing a dynamic process toward consumer self-realization and growth. Consumer autonomy is defined as "consumers' ability to make and enact decisions on their own, free from external influences imposed by other agents" (Wertenbroch et al., 2020, p. 430). Within the AI market contexts, we define consumer autonomy as "consumers' ability to make decisions and act independently without undue external influence of control or manipulation through data utilization."

This definition emphasizes consumers' internal freedom to make choices based on personal preferences, unimpeded by AI control or manipulation, while also highlighting the possibility of making informed decisions. In an AI-driven environment, it implies that consumers should be guaranteed the external freedom to act freely, ensuring both the freedom to act as they desire without external interference and the freedom to choose according to their wishes—conditions under which autonomy is ultimately realized (Graef et al., 2023).

2.2. Self-determination theory

Self-determination theory (SDT), a cornerstone theory regarding autonomy, is a comprehensive theory about human motivation. Central to this theory is the concept of autonomy, which SDT posits is crucial for human happiness, which hinges on the satisfaction of three basic psychological needs: autonomy, competence, and relatedness (Deci & Ryan, 2000).

SDT posits that humans are inherently active and growth-oriented organisms who are inclined to integrate their psychological needs into a cohesive self and into broader societal structures (Ryan & Deci, 2006). Autonomy is crucial as it involves aligning external influences with internal desires, making decisions and taking actions in a self-determined manner (Scherhorn, 1985). The theory emphasizes the profound impact that social environments can have on an individual's motivation and well-being. It also suggests that the characteristics of one's environment can either enable or constrain autonomy, making it a critical determinant of how individuals function within that context (Deci & Ryan, 2000).

In today's shopping context where AI-driven recommendations are prevalent, the environment can either permit free choice or drive consumers toward restricted options, thus significantly influencing consumer autonomy (Sankaran et al., 2021). SDT focuses on understanding the nature and outcomes of autonomy and how autonomy is developed, diminished, or facilitated by certain social conditions. Specifically, SDT emphasizes the interaction between an individual's intrinsic tendency toward integration and their vulnerability to manipulation (Ryan & Deci, 2006). Applying this theory to AI shopping experiences provides a deeper understanding of how AI-recommended systems may facilitate or restrict consumer autonomy.

SDT highlights three basic psychological needs: autonomy, competence, and relatedness. However, this study focuses exclusively on autonomy because in AI-enabled shopping environments, issues such as algorithmic opacity, personalization pressure, and restricted choices are more directly related to consumers' sense of self-determined decision-making than to competence or relatedness. Consistent with prior empirical applications of SDT in AI contexts (Ou & Li, 2025), we selected autonomy as the most

theoretically aligned and practically salient mediator for explaining consumer satisfaction in AI-mediated shopping experiences.

2.3. AI-based recommendation and consumer satisfaction

Various empirical studies across e-commerce, digital platforms, and fintech have shown that AI-driven recommendation systems can significantly enhance user satisfaction, particularly systems offering personalized suggestions. For instance, when recommendations align with users' goals, they evoke a "feeling right" response that boosts satisfaction, whereas mismatched recommendations lead to psychological reactance (He et al., 2024). Personalization and recommendation quality also enhance perceived usefulness and satisfaction, thereby encouraging continued use and favorable evaluations (Kim & Kim, 2025; Yin & Qiu, 2021). These effects have been observed across diverse sectors, including retail, grocery, and media platforms (Kim & Kim, 2025; Pramudito et al., 2021).

Researchers have also raised concerns about negative user experiences, particularly regarding opacity, privacy intrusion, and ethical manipulation. Opaque "black-box" algorithms can diminish users' perceived autonomy, leading to skepticism or resistance (Zarsky, 2016). Among these issues, privacy has been the most empirically studied; however, many concerns remain at the conceptual level. Although recent studies have begun to address perceived fairness and transparency (Wang et al., 2024), empirical evidence remains limited on how these negative perceptions influence satisfaction. To summarize these insights, Table 1 categorizes representative studies on positive and negative influences of AI recommendations, highlighting both the scope of current knowledge and existing gaps. This underscores the need to systematically investigate a broader set of negative factors—not just privacy-related factors—in shaping consumer responses to AI-driven recommendation systems.

3. Overview of the studies

This research employs a mixed-methods approach to explore and better understand the complex relationships among various constructs (Babu et al., 2024). This study combined qualitative analysis of interview themes with quantitative analysis through structural equation modeling (SEM).

The interview procedures and the questionnaire used in the survey were conducted following prior review and approval by the Institutional Review Board (IRB) at a university located in Seoul (IRB No. 2312/004-021).

Unlike prior studies that adopt predefined constructs to assess users' perceptions, this research begins with a qualitative phase to inductively capture the real-world complexity of users' AI-driven shopping experiences. Given that AI systems can induce unfamiliar psychological responses, it is important to uncover dimensions of user-derived perceptions before proceeding to scale development and quantitative validation. This approach ensures that the model reflects actual consumer sentiments and concerns, rather than imposing assumptions drawn from unrelated contexts.

In Study 1, a qualitative research approach was employed to explore consumer perceptions of AI-driven recommendations and their impact on decision making. This exploratory investigation identified both positive and negative perceptions, which were then developed into a scale. Study 2 examined the influence of the consumer perceptions derived from Study 1 on consumer satisfaction with their AI shopping experience. We also investigated whether consumer autonomy mediated this relationship. The overall findings of the research are summarized in Figure 1.

4. Study 1: Qualitative study

4.1. Data collection and data analysis

To examine consumer perceptions with decision-making experience facilitated by AI-driven recommendations, in-depth, one-on-one interviews were conducted with 20 consumers in South Korea. Participants were aged 20–50 with prior experience using platforms featuring AI-based

Table 1. Positive and negative influences on user satisfaction in AI recommendation systems.

	Study (year)	Focus/context	Methodology	Key findings
Positive influences	Yin and Qiu (2021)	AI marketing tech on shopping platforms	Empirical – Survey (SEM)	Accuracy, insight, and interactive experiences raise perceived value, which increases purchase intention.
	Pramudito et al. (2021)	E-grocery platform recommendation quality	Empirical – Survey (SEM)	E-satisfaction mediates the effect of platform quality (design, fulfillment, privacy, service) on e-loyalty.
	Noor et al. (2022)	AI service agents (AISA) service quality	Scale development – Qualitative interviews, and quantitative validation	Identified 6 AISAQUAL (efficiency, security, availability, enjoyment, contact, anthropomorphism), which enhance perceived value, satisfaction, and loyalty.
	He et al. (2024)	Recommendation fit with user shopping goals	Empirical – Experiment	Alignment between the recommendation type and user's goal increases satisfaction via "feeling right." A mismatch leads to psychological reactance.
	Kim and Kim (2025)	AI information/system quality and user satisfaction	Empirical – Survey	Evaluated impacts of information quality (completeness, accuracy and format) and system quality (reliability, flexibility and timeliness) on recommendation quality.
Negative influences	Mittelstadt et al. (2016)	Ethical risks in algorithmic decision-making	Conceptual – Framework Paper	Highlights fairness, accountability, and opacity as key ethical risks affecting user trust and system legitimacy.
	Zarsky (2016)	Algorithmic opacity and user autonomy	Conceptual – legal essay	Opaque AI systems undermine autonomy and can cause resistance, which calls for explainability and transparency.
	Susser et al. (2019)	Online manipulation via personalization	Conceptual – Tech ethics analysis	Warn that personalized content can covertly exploit user biases, undermining autonomy and trust, which calls for greater transparency and oversight to prevent manipulative personalization.
	Vaassen (2022)	Opacity and personal autonomy in AI	Conceptual – Philosophical analysis	Opacity limits users' understanding and undermines autonomy in decision-making.
	Wang et al. (2024)	Personalization–privacy paradox in retail	Empirical – Experiment	"Related product" formats reduce perceived privacy invasion and increase acceptance. "Similar product" formats raise more privacy concerns.

recommendation systems. They were selected considering relatively balanced gender and age diversity. A summary of the participants' demographics is presented in Table 2.

Participants were not shown a specific shopping platform but were encouraged to recall AI-driven recommendations from platforms they had personally used. Most participants mentioned major e-commerce platforms in South Korea such as Naver Shopping (a search-based shopping aggregator), Coupang (a leading platform known for fast delivery), and Zigzag (a mobile fashion commerce app popular among younger consumers). Participants were asked about their perceptions of experiences using AI recommendations that influenced their decision-making.

Given the diverse nature of consumer experiences in AI environments, this study adopted an inductive thematic analysis approach focusing on collecting data around themes and deriving themes directly from the data. Thematic analysis involves identifying, analyzing, and interpreting patterns of meaning within qualitative data (Clarke & Braun, 2017). Thematic analysis can employ both inductive, data-driven methods and deductive, theory-based approaches (Byrne, 2022). The essence of thematic analysis lies not merely in summarizing the data but also identifying and interpreting key features within the data based on the research questions (Clarke & Braun, 2017).

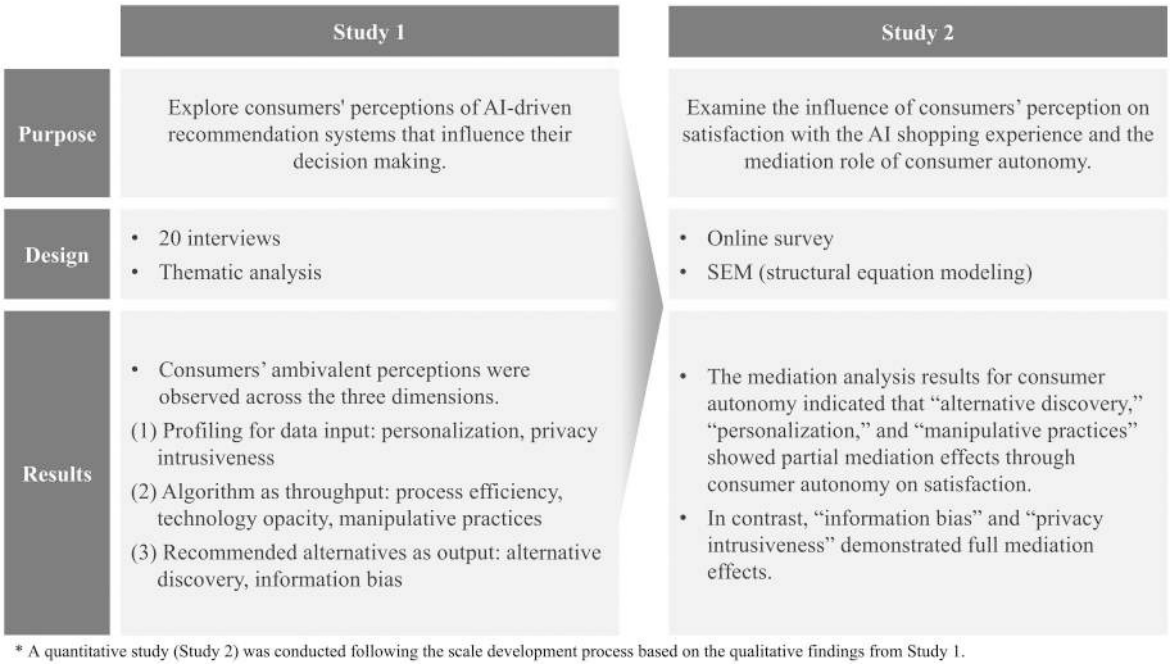


Figure 1. Overview of the studies.

Table 2. Demographic characteristics of participants in study 1.

ID	Age group	Gender	Occupation	ID	Age group	Gender	Occupation
A-1	20s	Male	College student	B-6	30s	Female	Office worker
A-2	20s	Male	Graduate student	C-1	40s	Male	Office worker
A-3	20s	Female	Graduate student	C-2	40s	Male	Office worker
A-4	20s	Female	Office worker	C-3	40s	Female	Office worker
A-5	20s	Female	Graduate student	C-4	40s	Female	Professional
B-1	30s	Male	Self-employed	C-5	40s	Female	Graduate student
B-2	30s	Male	Office worker	D-1	50s	Male	Office worker
B-3	30s	Male	Graduate student	D-2	50s	Male	Office worker
B-4	30s	Female	Office worker	D-3	50s	Female	Office worker
B-5	30s	Female	Public servant	D-4	50s	Female	Office worker

The thematic analysis process involves developing codes as the smallest unit of analysis that captures notable data characteristics related to the research questions and then deriving themes. This study analyzed interview transcripts based on the six-phase process proposed by Braun and Clarke (2006) and reviewed by Byrne (2022). The process includes (1) becoming familiar with the data, (2) generating initial codes, (3) generating themes, (4) reviewing potential themes, (5) defining and naming the themes, and (6) producing the report.

Following these steps, we repeatedly read the transcribed interview materials, generated initial codes, and categorized the collected codes into larger themes. The transcripts were analyzed using the qualitative data analysis software MAXQDA, which facilitates recording and aggregating codes and themes. This software enabled repeated checks of the transcriptions and themes throughout the analysis. Through iterative coding and categorization, we refined the data to derive insightful analytical results. Experienced researchers in consumer studies reviewed the results to ensure the validity and appropriateness of the analysis. To enhance the reliability of the study results, we cross-verified categories with existing academic and practical literature, integrating relevant reports to enhance understanding.

4.2. Study 1 results and discussion

The process of AI recommending products to consumers is divided into three stages. In the first stage, the system collects consumer data to build input profiles, focusing on their characteristics and preferences. This forms the foundation for understanding their needs and values. In the second stage, algorithms match this data with suitable products, using computational techniques to identify the best

recommendations. Finally, in the third stage, the system presents the recommended alternatives, delivering tailored options that align with consumer expectations.

The thematic analysis results in this study indicated that consumer perceptions of AI-driven recommendations are composed of three stages: (1) Profiling for data input, (2) Algorithm as throughput, and (3) Recommended alternatives as output. Ambivalent consumer responses were observed for each stage and the themes were derived across all three stages, which suggests that consumers experience both positive and negative perspectives of technology-based products and services, which is consistent with prior research (Yap et al., 2021).

4.2.1. Theme 1: Profiling for data input

The core value of AI-driven recommendations is in matching products that are aligned with each consumer's needs and preferences. AI technology integrates various data sources (e.g., consumers' purchase history, online activity, demographic characteristics, the behavior of peer groups) to accurately predict the consumer's needs and preferences. While these efforts provide personalized recommendations based on the consumer's preferences, they also increase concerns about privacy intrusiveness that consumers may not desire. As shown in Table 3, algorithms that learn consumers' preferences and predict the most favored products or services are generally seen as practical and convenient. However, when consumers wish to explore or reveal their tastes and personal characteristics through their choices, such services may lead to consumer discomfort. This finding aligns with previous research (André et al., 2018).

4.2.2. Theme 2: Algorithm as throughput

AI-driven recommendations use algorithms to enhance efficiency in the purchasing process. The benefits include reducing the time consumers spend exploring and comparing products, enabling easier comparison of alternatives, and minimizing cognitive dissonance in decision making. However, the impact of this efficiency is not always positive. Beneath the surface of efficiency lies the challenge of understanding how the technology operates, as well as concerns about possible unfair transactions that prioritize corporate profit. As shown in Table 4, these perceptions of efficiency and opacity lead consumers to both appreciate the convenience but also distrust the practices. Specifically, opacity includes both the operation of the technology and the manipulative practices of the seller.

4.2.3. Theme 3: Recommended alternatives as output

AI-driven recommendations give consumers easier access online to a wide variety of products and information. While alternative discovery (i.e., recommendations) broadens the scope of consumer choices and increases autonomy, it also leads to the "paradox of choice," where an expanded range of

Table 3. Theme 1: Profiling for data input.

Representative responses	First-order	Second-order
When I happen to have a need for a product, I feel happy and grateful when it shows me what I want.	Needs fulfillment	Personalization
I like how it shows me a mix of pants and skirts that would go well with a t-shirt, pairing them together.	Matching preferences	
It helps me narrow down the search steps in the direction I want to go, allowing me to make an ideal purchase.	Understanding requirements	
When there's a product category I'm interested in, and I see ads related to it or ads that align with what I wanted, I tend to click on them.	Interest consistency	
Once I agree to the use of my personal information, I feel like AI connects me to a flood of ads from various sources.	Misuse of personal information	Privacy intrusiveness
If sensitive information that I don't want to share with others is being used, it feels like I'm being made naked in front of AI, which makes me feel scared.	Potential use of sensitive information	
The reason I feel embarrassed about letting AI know my preferences is that I'm afraid my private tastes, which AI knows, might be exposed to others.	Concerns about revealing personal preferences	
Even if its capabilities improve, it still feels like being surveilled, which makes me uncomfortable.	Surveillance concerns	

options can complicate decision making (Schwartz, 2004). The abundant alternatives can increase opportunity costs and induce feelings of regret or anxiety (Steffel & Williams, 2018). Furthermore, AI algorithms, which predict consumer preferences based on past behaviors and provide personalized content, may create filter bubbles, leading to biased outcomes (Bjørlo et al., 2021). The in-depth consumer interviews revealed that consumers experience both discovery of alternatives and information bias, as shown in Table 5.

5. Study 2: Quantitative study

5.1. Model and hypothesis development

All consumer perception variables in this study were developed based on the findings of Study 1, and conceptually grounded in the unique characteristics of AI-driven recommendation systems. Constructs such as alternative discovery, process efficiency, and personalization reflect how AI technologies proactively infer consumer preferences through real-time behavioral data and present individualized options that go beyond explicit consumer input (Berente et al., 2021; Wertenbroch et al., 2020). These personalized mechanisms enhance the user experience in ways that are not typically observed in traditional rule-based systems.

Constructs such as technological opacity, information bias, privacy intrusiveness, and manipulative practices capture the distinctive risks embedded in AI's data-driven, automated personalization processes (Bjørlo et al., 2021; Susser et al., 2019). These concerns reflect AI-specific psychological responses such as uncertainty about algorithmic logic (Berente et al., 2021), perceived loss of control, and discomfort from hyper-targeted persuasion (Wertenbroch et al., 2020). Together, these variables represent experiential dimensions that are unique to AI-powered environments and are essential for understanding both the benefits and ethical concerns surrounding AI-based recommendations.

5.1.1. Positive perception toward AI-driven recommendation system

AI-driven recommendations enhance "alternative discovery" by providing consumers with a broader range of choices and information. This enables consumers to make more conscious and autonomous

Table 4. Theme 2: Algorithm as throughput.

Representative responses	First-order	Second-order
It saves me significant time by efficiently organizing and focusing on the information I need.	Saving time in exploration	Process efficiency
It's great because it makes it easy and quick to compare the alternatives	Ease of comparing alternatives	
AI reduces the difficulty of the decision-making process and helps me make decisions that I can be satisfied with.	Reduced options	
There are many tasks that are overwhelming and exhausting to search for individually. I think it's great that artificial intelligence takes care of the parts I find tedious.	Reduction of unnecessary hassles	
I wonder what it checks about me to keep showing me these things … Even when I'm talking with my husband, it collects all that information and shows it to me.	Opacity in data utilization	Technology opacity
If I search on the internet myself, I can verify the sources, but when AI finds it, I can't tell where it's pulling the information from or what the source is.	Opacity in the basis for results	
It's an information asymmetry. I don't think I can figure out where or how things went wrong.	Difficulty understanding the mechanism	
Even when artificial intelligence reduces the information or leaves only one option, I still think, "Can't I find better information than this myself?"	Skepticism about the best outcome	Manipulative practices
The fact that it shows me these results as recommendations for me gives me the feeling that it's leading me in one direction.	Manipulation of consumer behavior	
I feel like companies might be giving more weight to certain products behind the scenes, such as showing more of the items with higher advertising expenditures.	Recommendation of advertised products	
It reminds me of items I've seen before, or it shows related products that people who bought this item also purchased. Like a chain of items … I think this could encourage overspending or impulse buying.	Inducing overspending and impulse purchases	

Table 5. Theme 3: Recommended alternatives as output.

Representative responses	First-order	Second-order
After I searched for shampoo and showed interest in hair care, it recommended supplements, and I ended up buying them. At first, I hadn't thought about buying supplements.	Discovery of unrecognized needs	Alternative discovery
I get to know new brands, and since I can easily access new information, I think my perspective broadens in the process.	Exploration of new products	
It allows me to consider aspects I hadn't thought of and discover better products, which has been really helpful.	Acquisition of valuable product information	
I like that AI recommendations provide information about new arrivals and current trends.	Keeping up with the latest trends	
When AI just decides and shows me "This is the best for you," it feels like my choice is being infringed upon, and I feel restricted.	Reduced diversity of information	Information bias
Sometimes, even if it's the same product, I think, "Isn't there something new?" and I feel like the opportunity to explore other products decreases.	Reduced access to information	
I don't always want just one type of product, but I start to wonder if I'm only being exposed to a narrow range of results, which might be a bit limiting.	Recommendations based on existing patterns	
My preferences change, but since it's based only on past data, it feels like my current situation isn't being reflected.	Limitations in changing preferences	
I wonder if we might end up in a society where we consume without thinking … where everyone ends up wearing the same thing, just like me.	Homogenization of preferences	

decisions based on their lifestyles, consumption patterns, and values. The fundamental motivation behind seeking diversity is the desire for novelty, surprise, change, and complexity, all of which lead to satisfying intrinsic needs (Chinchanachokchai et al., 2021).

Algorithmic decision support significantly improves "process efficiency" by providing more accurate services and vital information for consumer decision making (Fan & Liu, 2022). As AI-driven recommendations streamline the decision-making process, reduce cognitive dissonance, and enable more efficient purchasing decisions, thereby enhancing decision satisfaction. Being able to shop with less effort also reduces the burden of the shopping process, thus allowing for more autonomous decisions (De Bellis & Johar, 2020).

"Personalization" in AI recommendations offers convenient options for online consumers, which helps alleviate information overload (Shen et al., 2023). Flaherty et al. (2019) showed that personalized recommendation technologies give consumers the freedom to choose products that align with their needs and interests. This freedom increases engagement and satisfaction with the app. These factors collectively suggest that positive perceptions of AI features can significantly influence consumer satisfaction.

H1: Positive perceptions related to (a) process efficiency, (b) alternative discovery and (c) personalization have a positive impact on consumers' satisfaction with their AI shopping experience.

5.1.2. Negative perceptions toward AI-driven recommendation system

Due to negative perceptions surrounding AI-driven recommendations, consumers may feel that their actions are being controlled by AI, which could adversely affect their sense of autonomy (Sankaran et al., 2021). Their negative perceptions of AI-driven recommendations, such as "information bias," can increase as algorithmic micro-targeting creates self-reinforcing cycles, heightening the prevalence of biased information (Wertenbroch et al., 2020). This echo chamber effect due to prolonged exposure to algorithmic recommendations exposes users only to homogeneous information, leading to excessive personalization, a narrower range of information, and biases. As a result, it can evoke negative emotions such as algorithm aversion (Lv et al., 2025). In addition, algorithms continuously recommend similar information to consumers' previous search preferences, which could result in redundant outcomes and limited exposure to broader categories (André et al., 2018). Thus, it diminishes consumer satisfaction due to unmet information needs (Zhang et al., 2024).

Concerns about "privacy intrusiveness" have intensified as products have become increasingly connected to the internet and as advertising technologies evolve to target consumers by tracking even offline behavior (Fassiaux, 2023). Emerging digital surveillance, characterized by extensive and persistent data collection, manipulates, predicts, and shapes consumer behaviors, which can severely impact consumers' freedom to shop and their overall satisfaction (Song & Lin, 2024).

"Technology opacity" of AI-based recommendation systems is another significant concern. The lack of transparency in how these systems operate, including their capabilities, performance, decision-making logic, and rationale, contributes to concerns about opacity (Luo et al., 2022). Consumers find it difficult to understand the operational mechanisms of AI, coupled with the lack of explanations regarding the principles or outcomes (Berente et al., 2021), which leads to information asymmetry and opacity concerns (Vaassen, 2022). The lack of transparency can also make it challenging for consumers to identify and rectify AI behaviors that deviate from their expectations.

Finally, "manipulative practices" in AI-driven recommendations manifest as opportunistic behaviors, where data-driven systems, while convenient and useful, also increase the likelihood of hidden interference and manipulation (Bjørlo et al., 2021). Such manipulations can exploit consumers' cognitive or emotional vulnerabilities, leading to choices that do not align with their initial or rational preferences (De Marcellis-Warin et al., 2022).

H2: Negative perceptions related to (a) information bias, (b) privacy intrusiveness, (c) technology opacity, and (d) manipulative practices have a negative impact on consumers' satisfaction with their AI shopping experience.

5.1.3. The mediation role of consumer autonomy

SDT posits that when the basic psychological need of autonomy is satisfied, it increases satisfaction. Although the decision-making process does not always yield optimal outcomes, the perception of autonomy in decision making plays a significant role in enhancing satisfaction and well-being (André et al., 2018). Individuals with fulfilled autonomy needs feel that their actions are voluntary and aligned with their own interests and values (Van Assche et al., 2018). For example, Xiong and Zhang (2024) examined the use of tourism app services and found that when consumers perceive autonomy, they are more likely to invest effort in their experience and concentrate on appreciating the specific service. These actions can enhance satisfaction. In contrast, consumers experiencing autonomy frustration may feel that their actions, thoughts, and emotions are being controlled, resulting in negative reactions (Van Assche et al., 2018).

H3: Consumer autonomy has a positive impact on consumers' satisfaction with their AI shopping experience.

Autonomy is primarily driven by intrinsic motivation, but it is also influenced by external events and environmental contexts. According to self-determination theory, environmental support is crucial for realizing autonomy (Deci & Ryan, 2000). Shen et al. (2023) underscored that experiencing autonomy during human-computer interaction (HCI) significantly affects consumer well-being. Their study found that recognizing the benefits of HCI can enhance autonomy, leading to increased self-control and well-being. Sankaran et al. (2021) suggested that the ideal conditions for enhancing autonomy include having sufficient choice and transparency, considering user preferences, and giving consumers the ability or assistance to make informed decisions. Their finding is closely aligned with the hypothesis of this study that alternative discovery, personalization, and process efficiency positively impact consumers' perceived autonomy.

AI-driven recommendations boost "process efficiency," allowing individuals to achieve goals with less effort and time, thus enhancing efficiency and providing opportunities for self-expansion (Zhang et al., 2024). Recommendations enable consumers to easily explore and compare various alternatives, increasing their opportunities to make choices based on their true preferences. The "alternative discovery" aspect suggests that when consumers perceive that they have ample choices, it enhances their autonomy. Husairi and Rossi (2024) emphasized that when AI-driven recommendation technologies meet consumers' diversity-seeking needs, it can lead to positive outcomes by actualizing choice

autonomy. Similarly, "personalization" through AI-driven recommendations enhances consumer autonomy. As consumers recognize higher levels of personalization, they gain more relevant knowledge, feel a stronger sense of self-expansion, and can make decisions based on their intrinsic preferences (Zhang et al., 2024).

H4: Consumer autonomy positively mediates the relationship between positive perceptions related to (a) process efficiency, (b) alternative discovery, and (c) personalization, and consumers' satisfaction with their AI shopping experience.

SDT posits that autonomy, defined as "self-governance," contrasts with heteronomy, which is influenced by external pressures such as rewards and punishments (Ryan & Deci, 2006). External control can also lead to adverse psychological effects. When AI decision making infringes on consumer autonomy, it often triggers psychological reactance, which reduces satisfaction (Fan & Liu, 2022). When AI dominates decision making, users may feel powerless, diminishing their adoption intentions (Frank & Otterbring, 2024).

In terms of "information bias," algorithmic micro-targeting may reinforce existing preferences and trap consumers in a "filter bubble" that limits exposure to diverse alternatives and leads to biased outcomes (Bjørlo et al., 2021; Wertenbroch et al., 2020). AI-driven recommendations, which present only a limited set of alternatives based on their algorithms, may act as a factor that weakens consumer freedom of choice (Fink et al., 2024). The perception of predetermined choices by AI erodes consumers' sense of control over decisions, undermining satisfaction (Lv & Huang, 2024). Privacy concerns also arise from the indiscriminate use of personal data for personalized recommendations, perceived as "privacy intrusiveness" (Wang et al., 2024). Lack of transparency in AI-driven recommendations, due to "technology opacity," hinders consumers from verifying the authenticity of the data, which affects their autonomy in decision making (Bjørlo et al., 2021; Vaassen, 2022). In addition, if AI systems prioritize company strategies over consumer preferences, driven by "manipulative practices," it may lead to a perceived restriction of consumer autonomy (Husairi & Rossi, 2024). Thus, understanding the impact of negative perceptions on autonomy is vital for enhancing consumers' AI experiences, as it plays a crucial role in mediating consumers' satisfaction with their AI shopping experiences.

H5: Consumer autonomy negatively mediates the relationship between negative perceptions related to (a) information bias, (b) privacy intrusiveness, (c) technology opacity, and (d) manipulative practices, and consumers' satisfaction with their AI shopping experience.

5.2. Measures development

To develop a scale measuring consumer perception of AI-driven recommendations, this study created various items based on our qualitative research findings and prior studies. The initial items comprised 28 statements across 7 factors. The sample was divided into two groups based on gender, with exploratory factor analysis conducted on Group 1, followed by two rounds of confirmatory factor analysis on Group 2 to validate the final scale. The consumer perception scale for AI-driven recommendations was finalized with both positive and negative perceptions. Finally, multi-group analysis was conducted to verify configural invariance and measurement invariance tests. The process of scale development and the final measurement items are shown in Supplementary Material.

This study measured consumer autonomy, a mediating variable, from the perspective of perceived autonomy in interactions with technology. A scale with five items developed by Sankaran et al. (2021) was revised and applied to align with the purpose of this study, including the item "AI-driven recommendations let the users do things their own way." Satisfaction with the AI shopping experience was measured by adapting items from Hu and Krishen (2019) and Hsieh et al. (2022), using four items. It included "I was satisfied with my experience of deciding which product option to choose" to fit the context of this study. Items were measured on a five-point scale (1 = very satisfied; 5 = very dissatisfied).

5.3. Sampling and questionnaire administration

In line with the demographic and conceptual scope of Study 1, we recruited participants aged 20–50 who had prior experience using AI-driven shopping platforms. According to Statista (2024), South Korea is one of the largest e-commerce markets in the world with 94 percent of the population using the internet. Respondents were allocated based on gender and age in proportion to the population demographics. After excluding responses deemed unsuitable for analysis due to inattentiveness, 560 participants were included in the study.

To enhance clarity and response consistency, the survey presented screenshots of AI-driven recommendations from three of the most widely used e-commerce platforms in South Korea: Naver Shopping, Coupang, and Zigzag. The screenshots illustrated various types of personalized recommendation mechanisms that are commonly encountered by users during online shopping such as similar product suggestions, history-based predictions, demographic-based recommendations, and real-time trending items. These examples helped respondents recall their experiences in a generalized AI recommendation context, rather than focusing on a specific platform. (See Supplementary Material A for the examples presented in the survey.)

To ensure the appropriateness of the study sample, a screening question was included, along with attention check questions in the middle of the survey. The demographic analysis of participants revealed a balanced gender distribution with males representing 52.14% ($n = 292$) and females 47.86% ($n = 268$). The sample was evenly divided across different age groups: under 20s at 25% ($n = 140$), 30s at 26.07% ($n = 146$), 40s at 24.29% ($n = 136$), and 50s at 24.64% ($n = 138$).

5.4. Reliability and validity of the measurement model

The measurement model was initially evaluated using confirmatory factor analysis (CFA), demonstrating good model fit ($\chi^2/df = 2.179$, CFI = 0.951, TLI = 0.943, RMSEA = 0.046, SRMR = 0.044). Next, we assessed the convergent and discriminant validity of the constructs. Convergent validity was confirmed by examining the composite reliability (CR) and average variance extracted (AVE). As shown in Table 6, the CR values for the constructs met the recommended threshold of 0.70, and the AVE values met the recommended threshold of 0.50. As indicated in Table 7, the square root of the AVE of each construct exceeded the corresponding inter-construct correlation values (Fornell & Larcker, 1981), indicating discriminant validity of the data.

5.5. Common method bias (CMB)

As a precautionary measure, attention-check questions were included to evaluate whether respondents were attentive when providing their answers. The confidentiality of their responses was guaranteed. Common method bias (CMB) was addressed using the statistical procedures recommended by Podsakoff et al. (2003). Harman's single-factor test revealed that a single factor explained only 32.39% of the total variance, which is significantly below the recommended threshold of 50%. The common latent factor (CLF) method (Podsakoff et al., 2003) was employed to assess CMB. Comparing estimates from the CLF model to those from the non-CLF model showed standardized estimate deviations ranging from 0.012 to 0.049. These findings suggest no significant CMB issues and confirm that including this construct enhances the predictive validity of the model.

5.6. Structural model

The findings of the structural model indicate that the positive perceptions of process efficiency (Effect = 0.261, $p < 0.001$), alternative discovery (Effect = 0.205, $p < 0.001$), and personalization (Effect = 0.227, $p = 0.003$) are significantly associated with consumers' satisfaction with the AI shopping experience. Details of the direct effects can be found in Table 8.

In terms of the negative perceptions, only manipulative practices (Effect = −0.105, $p = 0.022$) shows a significant negative direct effect on satisfaction, while information bias (Effect = −0.010, $p = 0.820$), privacy intrusiveness (Effect = −0.004, $p = 0.925$), and technological opacity (Effect = −0.019, $p = 0.573$)

Table 6. Measurement model.

Variables	Items		Factor loading	AVE	CR	Cronbach's α
Process efficiency	Using AI-driven recommendations …					
	PE1	… Takes less time to browse through alternatives.	0.788	0.577	0.804	0.815
	PE2	… Makes it easier to compare alternatives.	0.771			
	PE3	… Reduces many unnecessary hassles.	0.719			
Alternative discovery	AI-driven recommendations help me …					
	AE1	… Discover needs I was not aware of.	0.665	0.545	0.781	0.772
	AE2	… Discover new products.	0.774			
	AE3	… Discover good products during the search process.	0.770			
Personalization	I feel AI-driven recommendations …					
	PN1	… Reflect my needs when making suggestions.	0.838	0.600	0.856	0.855
	PN2	… Consider my needs as preferences when making recommendations.	0.806			
	PN3	… Know what I want when making recommendations.	0.757			
	PN4	… Match my interests when making recommendations.	0.689			
Information bias	AI-driven recommendations …					
	IB1	… Lessen information diversity.	0.848	0.635	0.836	0.821
	IB2	… Narrow my access to information.	0.881			
	IB3	… Expose me to information based on existing patterns.	0.639			
Privacy intrusiveness	When I use AI-driven recommendations …					
	PI1	… I am afraid that companies might misuse my personal data.	0.906	0.728	0.914	0.911
	PI2	… I am afraid of disclosing sensitive private information.	0.940			
	PI3	… I am afraid that preferences I don't want to reveal might be exposed.	0.741			
	PI4	… I feel I am under surveillance.	0.811			
Technology opacity	It isn't clear to me …					
	TO1	… What kind of data AI uses to generate recommendations.	0.830	0.724	0.887	0.884
	TO2	… Why the items are recommended to me.	0.919			
	TO3	… How AI-driven recommendations decisions are made.	0.800			
Manipulative practices	I feel that AI-driven recommendations …					
	MP1	… Are inclined more to merchants' interest than to my interests.	0.702	0.534	0.820	0.817
	MP2	… try to persuade me to act in a certain way.	0.817			
	MP3	… Are biased toward the merchants' product promotion interests, rather than my preferences.	0.752			
	MP4	… Encourage overspending and impulsive buying.	0.640			
Consumer autonomy	AI-driven recommendations …					
	CA1	… Let me act how I wanted rather than because I was told to.	0.695	0.594	0.879	0.876
	CA2	… Let me be in control of what I do.	0.832			
	CA3	… Let me do things my own way.	0.804			
	CA4	… let me make my own decisions.	0.819			
	CA5	… Provide choices based on my true interests.	0.690			
Satisfaction with the AI shopping experience	SE1	Overall, I am satisfied with the experience of using AI-driven recommendations.	0.836	0.615	0.864	0.858
	SE2	I think using AI-driven recommendations is a good idea.	0.747			
	SE3	I am satisfied with my experience of deciding which product option to choose.	0.841			
	SE4	I find the process of deciding which product to buy interesting.	0.704			

Abbreviations: AVE: average variance explained; CR: composite reliability.

Table 7. Discriminant validity.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1)	***0.760***								
(2)	0.518	***0.738***							
(3)	0.675	0.585	***0.775***						
(4)	−0.129	−0.310	−0.276	***0.797***					
(5)	−0.197	−0.233	−0.232	0.479	***0.853***				
(6)	−0.203	−0.155	−0.195	0.207	0.343	***0.851***			
(7)	−0.276	−0.290	−0.332	0.429	0.496	0.371	***0.731***		
(8)	0.386	0.449	0.505	−0.482	−0.481	−0.272	−0.506	***0.771***	
(9)	0.612	0.606	0.675	−0.360	−0.354	−0.240	−0.450	0.632	***0.784***

Note: Bold diagonal values show the square root of the AVE, and italicized values indicate correlations. AVE: diagonal values indicate the square root of AVE of the individual latent constructs.
(1) Process efficiency, (2) alternative discovery, (3) personalization, (4) information bias, (5) privacy intrusiveness, (6) technology opacity, (7) manipulative practices, (8) consumer autonomy, (9) satisfaction with the AI shopping experience.

are not significant direct predictors of satisfaction. Therefore, H1a, H1b, H1c, and H2d are supported, while H2a, H2b, and H2c are not supported.

The impact of consumer autonomy on satisfaction (H3) is significant, with consumer autonomy showing a strong positive effect on satisfaction (Effect = 0.295, $p < 0.001$), the strongest among all variables in the model.

In terms of the impact on consumer autonomy, process efficiency (Effect = −0.016, $p = 0.855$) is not significant, whereas alternative discovery (Effect = 0.156, $p = 0.019$) and personalization (Effect = 0.243, $p = 0.009$) are positively associated with consumer autonomy. Information bias (Effect = −0.213, $p < 0.001$), privacy intrusiveness (Effect = −0.159, $p < 0.001$), and manipulative practices (Effect = −0.233, $p < 0.001$) have negative impacts on consumer autonomy, while technology opacity (Effect = −0.014, $p = 0.730$) is not significant. Therefore, H4b, H4c, H5a, H5b, and H5d are supported, while H4a and H5c are not supported.

We tested the structural paths from consumer perception variables to satisfaction and autonomy were tested, as shown in Figure 2. Hypotheses H1 and H2 focused on the direct effects of consumer perceptions on satisfaction, whereas H4 and H5 examined whether these effects were mediated by consumer autonomy. To provide a more comprehensive evaluation of these relationships, we conducted a mediation analysis. The results revealed both direct and indirect effects (Table 9). Some perception constructs (e.g., information bias, privacy intrusiveness) did not show significant direct effects on satisfaction but had significant indirect effects via consumer autonomy. This finding confirms that autonomy mediates the relationship between the perception variables and satisfaction. By modeling both direct and mediated paths, we provide a more nuanced understanding of how AI-based recommendation systems influence consumer satisfaction.

Standardized estimates of direct, indirect, and total effects were analyzed to capture the mediating role of consumer autonomy (Table 9). The R^2 value for consumer autonomy is 0.547 and consumers' satisfaction with the AI shopping experience is 0.789. The structural model returned good fit indices (CMIN/DF = 2.180 ($p < 0.001$), CFI = 0.951, TLI = 0.943, RMSEA = 0.046, SRMR = 0.044) (Hair et al., 2010).

6. Results

The results of this study highlight several important conclusions. Study 1 investigated consumers' decision-making experiences with AI-driven recommendations through in-depth interviews. Consumer perceptions of these systems were first explored across three dimensions: "profiling for data input," "algorithm as throughput," and "recommended alternatives as output." Positive and negative perceptions were identified in each dimension and subsequently developed into a scale.

In the "profiling for data input" aspect, the results indicated that AI-driven recommendations enhance consumer satisfaction and streamline the shopping experience by offering product recommendations that align with consumer preferences. However, concerns over excessive personal data collection lead to negative perceptions of privacy intrusiveness.

Regarding "algorithm as throughput," AI systems reduce the hassle of searching and shorten the time consumers spend finding desired products, thus enhancing process efficiency. However, consumers distrust the opacity of the technology. They find it difficult to understand why certain products are recommended, making them suspicions of technology opacity. Participants were also concerned that AI systems might manipulate consumer behavior toward impulse buying and overconsumption, indicating potential manipulative practices.

In the "recommended alternatives as output" dimension, AI-driven recommendations helped consumers discover new interests and needs, offering unexpected product suggestions which benefited alternative discovery. However, they perceived that the information bias inherent in AI technology limited choice variety.

Study 2 examined whether consumer autonomy mediates the relationship between consumer perceptions and satisfaction. SEM showed that "information bias" and "privacy intrusiveness" significantly impact satisfaction only when mediated by consumer autonomy, demonstrating a complete mediation effect. Given the substantial impact of consumer autonomy on satisfaction, "information bias" and "privacy intrusiveness" are crucial in determining satisfaction through autonomy.

These findings indicate that "alternative discovery," "personalization," and "manipulative practices" have significant effects on both consumer autonomy and satisfaction, indicating partial mediation. Consumers enhance their autonomy and satisfaction when they have diverse options that align with

Table 8. Direct relationships.

Hypotheses		Relationship	β	SE	*t*-Value	ρ
H1	H1a	Process efficiency → satisfaction	0.261***	0.065	3.501	0.000
	H1b	Alternative discovery → satisfaction	0.205***	0.060	3.684	0.000
	H1c	Personalization → satisfaction	0.227**	0.068	2.931	0.003
H2	H2a	Information bias → satisfaction	−0.010	0.029	−0.227	0.820
	H2b	Privacy intrusiveness → satisfaction	−0.004	0.021	−0.094	0.925
	H2c	Technology opacity → satisfaction	0.019	0.024	0.563	0.573
	H2d	Manipulative practices → satisfaction	−0.105*	0.040	−2.285	0.022
H3		Consumer autonomy → satisfaction	0.295***	0.037	60.041	.000
H4	H4a	Process efficiency → consumer autonomy	−0.016	0.102	−0.183	0.855
	H4b	Alternative discovery → consumer autonomy	0.156*	0.093	2.356	0.018
	H4c	Personalization → consumer autonomy	0.243**	0.106	2.616	0.009
H5	H5a	Information bias → consumer autonomy	−0.213***	0.046	−4.133	0.000
	H5b	Privacy intrusiveness → consumer autonomy	−0.159***	0.034	−3.433	0.000
	H5c	Technology opacity → consumer autonomy	−0.014	0.039	−0.345	0.730
	H5d	Manipulative practices → consumer autonomy	−0.233***	0.063	−4.214	0.000

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

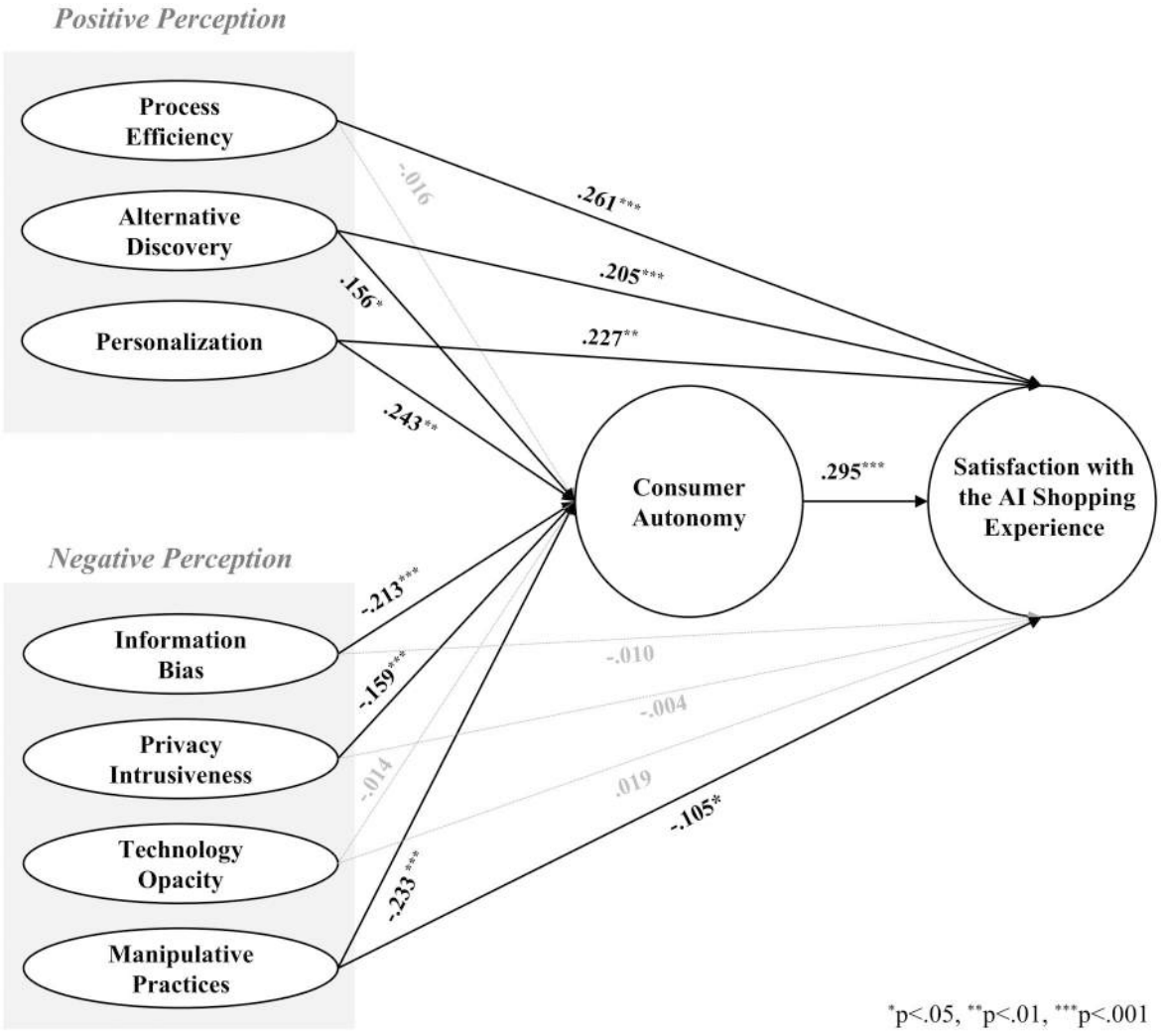


Figure 2. Results of study 2.

their preferences. In contrast, "manipulative practices" negatively affect both autonomy and satisfaction. This research also empirically confirms that consumer autonomy plays a critical linking role in how perceptions of AI-driven recommendations translate into consumers' satisfaction with their AI shopping experience. These findings suggest that AI-driven recommendation systems should prioritize a consumer-centric design to realize consumer autonomy.

Several hypothesized relationships are not supported. Specifically, process efficiency does not significantly predict consumer autonomy, and technological opacity, information bias, and privacy intrusiveness do not show significant direct effects on satisfaction. These findings and their possible explanations are further discussed in the following section.

7. Discussion and conclusion

7.1. Theoretical implications

This study makes several theoretical contributions to the literature. First, this study is significant in terms of empirically analyzing the role of consumer autonomy, which has been discussed mostly in theoretical terms. This study comprehensively examines the impact of AI-driven recommendations on satisfaction and addresses both consumer expectations and concerns that influence consumer autonomy. Unlike previous studies that have treated autonomy as a peripheral concept or a normative ideal, this research empirically positions it as a core explanatory mechanism in understanding user responses to AI systems.

Table 9. Direct, indirect, and total effects of consumer perceptions on satisfaction via consumer autonomy.

Relationship			Bootstrapping 95% CI			Direct effect	Indirect effect	Total effect
			LLCI	ULCI	*p*			
H4	H4a	Process efficiency → Consumer autonomy → satisfaction	−0.070	0.062	0.902	0.261***	−0.005	0.256
	H4b	Alternative discovery → Consumer autonomy → satisfaction	0.006	0.104	0.029	0.205***	0.046*	0.251
	H4c	Personalization → Consumer autonomy → satisfaction	0.012	0.142	0.013	0.227**	0.072*	0.299
H5	H5a	Information bias → Consumer autonomy → satisfaction	−0.114	−0.026	0.001	−0.010	−0.063**	−0.073
	H5b	privacy intrusiveness → consumer autonomy → satisfaction	−0.083	−0.016	0.002	−0.004	−0.047**	−0.051
	H5c	Technology opacity → Consumer autonomy → satisfaction	−0.031	0.022	0.785	0.019	−0.004	0.015
	H5d	Manipulative practices → Consumer autonomy → satisfaction	−0.124	−0.031	0.002	−0.105*	−0.069**	−0.174

Note 1: All estimates are standardized and significant at the 95% level: bootstrap iterations = 1000 using the bias-corrected percentile bootstrap method. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Second, while consumer autonomy has often been perceived as a value that may be compromised in a technological environment, this research verifies through mediation effects that technological benefits can enhance the perception of consumer autonomy and increase satisfaction. This finding suggests that AI recommendation technology could be an effective tool to enhance consumer autonomy. This result challenges the conventional view that technology and autonomy are in conflict, and instead demonstrates that AI features like personalization and alternative expansion can promote user agency.

Third, this study aimed to emphasize the significance of consumer autonomy by addressing the effects of positive perceptions as well as negative perceptions, which have typically been overlooked in benefit-focused technological advancements. Topics such as AI ethics, fairness, and privacy are explored in depth. These topics are highly relevant in the current era, yet they have often been neglected in the research (Mariani et al., 2022). Negative factors such as information bias, privacy intrusiveness, and manipulative practices have traditionally been discussed only in ethical terms and are often considered irrelevant to satisfaction, which is a practical outcome for businesses. This research reevaluates the importance of these negative elements by analyzing their impact on satisfaction mediated through consumer autonomy. It demonstrates that technological advancements should not only focus on enhancing positive aspects but also effectively manage negative aspects to respect and protect consumer autonomy. This dual focus necessitates appropriate operational strategies. By empirically testing the behavioral consequences of underexplored negative perceptions, this study fills a critical gap in the literature. To date, studies have been largely conceptual in their treatment of such concerns.

7.2. Practical implications

There are several practical implications based on the results of this study. First, the results emphasize the need for technological alternatives that enhance consumer autonomy in AI-driven recommendations. Current algorithms often deliver unilateral recommendations and use consumer data in ways that limit consumers' control over their personal information. It is vital to provide technological support enabling consumers to manage their data and explore items autonomously. Practical interface solutions include mechanisms for adjusting algorithm settings based on consumer preferences and developing tools for proactively managing privacy and empowering consumers to control their data (Calvo et al., 2020). Generative dialog-based systems enhance autonomy by allowing consumers to input specific requirements and receive tailored results. Additional features should allow consumers to verify and adjust algorithm-derived preferences, ensuring that consumer autonomy is respected.

Second, establishing a consumer-centric technological framework ensures that consumers actively participate in the purchasing process. Recent discussions on the "human-in-the-loop" principle have advocated for increased human involvement in AI decisions (Bjørlo et al., 2021), suggesting that the recommendations be expanded beyond expert intervention to direct consumer input. Stronger principles to prevent unfair practices and indiscriminate data use must also be established and implemented in a way that ensures consumer autonomy.

Third, corporate self-regulation is crucial in AI-dominated environments. Businesses must rigorously manage data protection and security, assuring consumers of the safety of their information, which can enhance their trust in AI systems. This commitment requires clear procedures that help consumers understand how their data are used and allowing them to make informed decisions independently. Incorporating consumer feedback allows consumers to select preferred attributes and compare alternatives, which fosters perceived autonomy.

Lastly, consumers play a crucial role in achieving autonomy. The findings indicate that efficiency of AI processes affects satisfaction. However, process efficiency does not significantly affect consumer autonomy. These results suggest that convenience alone may not make consumers feel autonomous. In fast-paced automated decision-making, consumers may not have enough opportunity to reflect on their true preferences and values. Consumers are not passive recipients of AI recommendations but should actively engage with AI systems to assert their autonomy (Fassiaux, 2023).

Several hypothesized effects are not supported in this research, which offers further insight into the role of consumer autonomy. We found that information bias and privacy intrusiveness do not have a direct impact on satisfaction, but both factors significantly undermine consumer autonomy. These

results suggest that the effects of negative perceptions may remain unnoticed, which confirms previous studies that have focused solely on satisfaction (Noor et al., 2022; Pramudito et al., 2021; Yin & Qiu, 2021). However, by identifying consumer autonomy as a key mediating variable, this study reveals that such perceptions can meaningfully influence satisfaction through indirect pathways.

Despite the emphasis on AI transparency in discussions on consumer digital literacy, this study found that technological opacity has no significant impact on consumer autonomy or satisfaction. Unlike other negative factors that directly impact consumers, technological opacity is less noticeable. While transparency issues may not be critical in everyday AI applications like shopping and media consumption, they become significant in high-stakes automated decision-making (Maclure, 2021). Additionally, explicitly revealing algorithmic processes can sometimes increase consumer aversion (Chen et al., 2024). Therefore, improving AI recommendation transparency requires a cautious and practical approach.

7.3. Limitations and future research avenues

This study emphasizes the role of consumer autonomy in enhancing satisfaction with AI-driven recommendations by measuring the psychological perception of behavioral possibilities related to consumer autonomy. Future research should build on our findings and measure the actual behavioral responses to consumer autonomy. Examining direct consumer autonomy behaviors rather than only reporting perceptions that the technological environment enables autonomy could provide a more accurate assessment of the actual levels of consumer autonomy. In addition, further discussion is necessary on specific strategies to realize consumer autonomy. Ongoing discussions on practical measures are also essential based on the findings of this study indicating the need for a multifaceted approach involving personal information management, control, and the freedom to explore and choose information.

Research should also be expanded to concretize ethical standards and principles to address consumers' negative perceptions. To realize consumer autonomy, it is crucial to effectively manage negative perceptions and develop concrete standards and implementation measures through empirical research. Specifically, researchers should focus on developing tools and procedures that enable effective management and control of personal data to ensure that consumers can use their data with confidence.

While this study incorporates both general and platform-specific user perceptions, the findings may be influenced by the structural and experiential features of the selected platforms. Given that AI recommendation systems differ in algorithmic architecture, interface design, and user interaction models, future research could conduct comparative analyses across diverse platforms or international markets to assess the generalizability and contextual variance of user perceptions and satisfaction drivers.

Based on this study, it would be feasible to apply and compare various evolving AI technologies. Generative AI is anticipated to be increasingly utilized in shopping platforms, led by companies like Amazon. Thus, it would be insightful to examine whether the realization levels of consumer autonomy differ in advanced technological environments considering the characteristics of emerging technologies.

This research highlights the various aspects and importance of consumer autonomy in AI-driven recommendation system environments. However, the manifestation of these aspects can vary depending on the AI-driven recommendation system. Therefore, consumer autonomy can be better understood by confirming how the perception of consumer autonomy changes across different purchasing situations where AI-driven recommendations are used.

Acknowledgement

This article is based on part of the first author's doctoral dissertation research. The first author gratefully acknowledges the financial support provided by the Department of Consumer Science at Seoul National University.

Author contributions

CRediT: **Eunsun Cho**: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing; **Jong-Youn Rha**: Conceptualization, Project administration, Supervision, Validation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Department of Consumer Science, Seoul National University.

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